Absolutely — below is a **clean, self-contained README.md** you can drop into your project and archive.
It's written so that **future you (or any AI you chat with later)** can immediately understand **what this project is**, **where it stopped**, and **how to continue**.

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# CoCo Automation Prototype (V1)

## Purpose

This project is a **low-cost prototype** to automate **Comparable Company (CoCo) analysis** using AI for KPMG Financial Instruments vluation Team, without relying on company-funded Azure AI or enterprise infrastructure.

The goal is to:

- Reduce manual effort in assessing CoCo strength
- Improve **consistency and explainability** of Strong / Median / Weak classifications
- Produce a **demo-ready workflow** to justify future funding and approvals

This is **not** intended to fully automate CapIQ or replace analyst judgment.
It is an **AI-assisted classification and reasoning tool**.

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## High-Level Workflow (V1)

### Step 1 — Manual (unchanged)

- Analyst runs **CapIQ Quick Search**
- Exports first-cut comparable companies to Excel
  ✅ No scraping, no automation of CapIQ

### Step 2 — Data Cleaning (semi-automated)

- CapIQ Excel is treated as **raw input**
- Only key fields are retained:
  - Company name, ticker
  - Country
  - Sector / industry
  - Short (or trimmed) business description
  - Revenue / market cap (if available)
- Noise (strategy, history, long narratives, Excel artifacts) is removed
- Cleaned data is exported to **JSON** for AI processing

### Step 3 — AI Comparison (core logic)

- Each CoCo is compared **one-by-one** against the target company
- Inputs to AI:
    - Target company profile (JSON)
    - Explicit scoring rules / rubric (JSON)
    - One cleaned CoCo record
- Outputs per company:
    - CoCo Level: **Strong / Median / Weak**
    - Numerical score (0–100)
    - Short, consistent reasons
    - Key differences
    - Confidence score
    - Flags for missing data

> AI is used for **classification + explanation**, guided by explicit rules.

### Step 4 — Output (V1 simple)

- Results are written back to:
    - coco_scored.json (source of truth)
    - coco_output.xlsx (for analysts / sharing)
- Formatting and VBA integration are **optional and deferred** to V2/V3

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## Design Principles

- ✅ **Excel at the edges, JSON in the middle**
- ✅ **Rules-driven**, not “AI vibes”
- ✅ **One company at a time** (predictable cost, easy debugging)
- ✅ **Manual CapIQ step preserved** (compliance-safe)
- ✅ **Low cost, personal API keys or local models**
- ✅ **Explainable and auditable outputs**

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## What This Prototype Does Well

- Standardizes CoCo assessment logic
- Produces consistent reasoning across analysts
- Separates **data**, **rules**, **prompts**, and **code**
- Is easy to demo and easy to extend

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## What This Prototype Does NOT Do (by design)

- ❌ No CapIQ scraping or automation
- ❌ No Azure / enterprise AI dependencies
- ❌ No RAG or document databases (yet)
- ❌ No auto-approval or deal decisions
- ❌ No guarantee of “correct” comps — analyst review required

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## Project Structure (V1)

```
coco-prototype/
│
├─ data/
│   ├─ raw/
│   │   └─ capiq_raw.xlsx
│   │
│   ├─ cleaned/
│   │   ├─ coco_candidates_clean.json
│   │   └─ target_clean.json
│   │
│   └─ output/
│       ├─ coco_scored.json
│       └─ coco_output.xlsx
│
├─ rules/
│   └─ coco_scoring_rules.json
│
├─ prompts/
│   └─ coco_scoring_prompt.txt
│
├─ src/
│   ├─ run_pipeline.py
│   ├─ excel_to_json.py
│   ├─ clean_descriptions.py
│   ├─ coco_scoring.py
│   ├─ json_to_excel.py
│   └─ llm_client.py
│
├─ vba/
│   └─ run_coco_analysis.bas
```

```
│
├─ config.json
└─ README.md
```

## Current Status

✅ V1 architecture defined
✅ CapIQ sample data reviewed
✅ Cleaning strategy decided
✅ JSON-first AI workflow agreed
✅ File structure finalized

🚧 Not yet implemented:

- Full Python scripts
- Final scoring prompt
- VBA automation (optional)

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## Next Logical Steps (Pick One)

1. Finalize **JSON schemas** for:
    - target_clean.json
    - coco_candidates_clean.json
2. Write **V1 scoring prompt** (strict JSON output)
3. Implement **run_pipeline.py** orchestration
4. Add **VBA one-click runner** (V2)
5. Extend to:
    - web context
    - historical CoCo benchmarks
    - RAG (V3)

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## Key Message for Future AI / Reviewers

> This project is a **controlled, explainable AI prototype** for CoCo analysis.
> It prioritizes **consistency, auditability, and analyst trust** over automation hype.

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If you want, next time you come back you can simply say:

> "I'm continuing the CoCo automation prototype — V1 JSON-based, rule-driven CoCo scoring from CapIQ exports."

…and any AI should be able to pick up **exactly** from here.

If you'd like, I can also:

- tighten this README for **manager review**, or
- add a short **"Why this matters"** section for funding approval.